

Durgesh Kumar

About

Hello, I am Durgesh. I study category theory and quantum foundations.

I recently completed my master's in mathematics at the University of Calgary, supervised by [Robin Cockett](#). My master's thesis explored a generalisation of the correspondence between orthonormal bases and copyable elements of Frobenius algebras to the infinite-dimensional setting. Before that, I studied mathematics and physics at Ashoka University. In winter 2026, I visited the [Compositional Systems and Methods](#) and [Logic and Semantics](#) groups at TalTech.

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Education

University of Calgary

Jan 2024 – Apr 2026

M.Sc. in Mathematics (thesis-based); Calgary, Canada
CGPA: 4.00/4.00. Supervisor: Robin Cockett.

Ashoka University

Aug 2021 – May 2022

Postgraduate Diploma in Research; Sonipat, India
Advanced major in mathematics; minor in physics.
GPA: 3.76; magna cum laude.

Ashoka University

Aug 2018 – May 2021

B.Sc. in Mathematics; Sonipat, India
Concentration in physics.

Research experience

Visiting Researcher

Winter 2026

Tallinn University of Technology (TalTech); Tallinn, Estonia
[Compositional Systems and Methods](#) and [Logic and Semantics](#) groups.

ACT Adjoint School participant

Feb - Aug 2023

University of Maryland, College Park; United States
Project: Message Passing Logic for Categorical Quantum Mechanics.
Supervisor: Priyaa Varshinee Srinivasan (NIST).

Research Assistant

Jun 2022 - Dec 2023

Ashoka University; India
Project: Monoidal Categories and Computation.
Supervisor: Debayan Gupta.

Publications

Unitary, Inner Product, and Dagger Categories

2026

Robin Cockett, Durgesh Kumar, and Priyaa Varshinee Srinivasan.
[Applied Category Theory \(ACT\) 2026](#).

Accepted to the 9th International Conference on Applied Category Theory, Tallinn, Estonia, July 2026.

Master's thesis

An adjunction between finiteness spaces and algebras in Lefschetz spaces

2026

Durgesh Kumar. Master's thesis in mathematics, University of Calgary.
Supervisor: Robin Cockett.
Nominated by the thesis committee for a national award.

Research in progress

We are working on generalising the notion of a non-stationary dagger developed in our paper to chiral settings. Our main aim is to connect it to the literature on linearly distributive categories and thus to categorical semantics of quantum mechanics. Further work will explore applications of this framework to modelling concurrent quantum protocols.

Teaching experience

Graduate Teaching Assistant, University of Calgary

MATH 271	Summer 2025
MATH 271, MATH 311	Winter 2025
MATH 271	Fall 2024
MATH 267, MATH 211	Winter 2024

Outreach and exposition

Linear Actegories September 2023

Paige Frederick, Durgesh Kumar, and Fabian Wiesner.
n-Category Café. We wrote an expository blog post on linear actegories as part of the ACT Adjoint School.

Applied Category Theory 2022

Ashoka University. I designed and co-taught a study module in applied category theory.

Course materials

For my blogs on formalisation and AI safety, see Mathematics and AI projects.

Talks, seminars, and posters

An adjunction between finiteness spaces and algebras in Lefschetz spaces March 2026

Talk. TallCat Seminar; TalTech, Tallinn, Estonia.

Comparative study of different formalisms of contextuality Fall 2025

Semester-long seminar. Quantum Group seminar series; University of Calgary.

Sheaf-theoretic view of contextuality September 2025

Talk. Peripatetic Seminar; University of Calgary.

Categorical semantics of infinite-dimensional quantum theory August 2025

Poster. International Quantum Horizons Symposium; Banff. | [Poster \(PDF\)](#)

A study of Lefschetz spaces June 2025

Talk. FMCS 2025; University of Ottawa. | [Slides \(PDF\)](#)

Towards Frobenius algebras in infinite-dimensional quantum mechanics May 2025

Talk. Alberta Mathematical Dialogues; University of Calgary. | [Slides \(PDF\)](#)

Towards Frobenius algebras in infinite-dimensional quantum mechanics February 2025

Talk. Peripatetic Seminar; University of Calgary.

The category of Lagrangian relations July 2024

Talk. FMCS 2024; Kananaskis. | [Slides \(PDF\)](#)

Honors and awards

Master's thesis nomination 2026

Nominated by the thesis committee for a national award.

Mitacs Globalink Research Award - Quantum Stream 2026

Tallinn University of Technology.

Mitacs Globalink Research Award - Quantum Stream

2025

University of Edinburgh; withdrawn because ATAS clearance was not received in time.

Dean's List

2020, 2022

Ashoka University; Monsoon 2020 and Spring 2022.

Ignite National Award for Grassroot Innovations

2012

Department of Science and Technology, Government of India.

Mathematics and AI projects

Heisenberg meets Anju

Notion of agents in quantum foundations and artificial intelligence

This post explores how the notion of agents in extended Wigner's friend scenarios in quantum foundations can add nuance to agentic frameworks in AI safety.

In programmes we trust: A categorical perspective

A (very) short sketch of the Curry-Howard-Lambek isomorphism

This post explores the trustworthiness of Lean certifications of AI-assisted (and AI-driven!) mathematics by giving a categorical account of the Curry-Howard-Lambek correspondence.

Inverse Galois Problem

We (Team Dirac) participated in the Inverse Galois Problem Competition, organised by the SAIR Foundation, and finished in 14th place.

[Project write-up](#) | [Leaderboard](#) | [Code](#)

LemmaPortfolio

We developed a benchmark for selecting a small shared set of useful lemmas for multiple Lean goals. This work is part of our larger motivation to study how language models can assist a mathematician in theory building. The best tested GPT-6 Astra runs found an optimal three-lemma portfolio on only 50% of test episodes (30/60).

[Project and code](#) | [Benchmark results](#)

what's my sheaf?

A global section calculator for empirical models based on Abramsky and Brandenburger's sheaf-theoretic formalism of contextuality.

A Quantum Latin Square of Order Six with Cardinality 29

Aishwarya P. Das and Durgesh Kumar.

We constructed a quantum Latin square of order six with cardinality 29, completing the cardinality spectrum for order six.

Ongoing work

We (Team Dirac) are currently working on the [Andrews-Curtis Conjecture \(ACC\) Challenge](#), organised by the SAIR Foundation, and hovering around the top five across the competition categories ([leaderboard](#)). We also have an ongoing project on formalising some results in mathematical general relativity in Lean ([repository](#)), and a project exploring some computational evidence in the Indus script ([repository](#)).

Patent

System to Integrate Driving License Validation in Vehicular Control

Patent record: 3408/DEL/2012.

Relevant coursework

Measure Theory; Algebraic Topology; Functional Analysis; Elliptic Curves; Algebraic Number Theory; Fourier Series; General Relativity and Cosmology; Mathematical Logic; Topics in Quantum Computation and Quantum Information; Category Theory.